MACHINE LEARNING ON RANDOM FOREST REGRESSION ASSIGNMENT

CHEBET

2026-10-14

#Setting a working directory

```
 setwd("C:/Users/Administrator/Desktop/PROFFESIONAL TRAINING")
```

#Load required libraries

```
library(dplyr)
```

```

Attaching package: 'dplyr'
```

```
The following objects are masked from 'package:stats':

    filter, lag
```

```
The following objects are masked from 'package:base':

    intersect, setdiff, setequal, union
```

```
library(tidyverse)
```

```
── Attaching core tidyverse packages ──────────────────────── tidyvers
e 2.0.0 ──
✔ forcats   1.0.1     ✔ readr     2.2.0
✔ ggplot2   4.0.3     ✔ stringr   1.6.0
✔ lubridate 1.9.5     ✔ tibble    3.3.1
✔ purrr     1.2.2     ✔ tidyr     1.3.2
```

```
── Conflicts ─────────────────────────────────────── tidyverse_conf
licts() ──
✖ dplyr::filter() masks stats::filter()
✖ dplyr::lag()    masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force
all conflicts to become errors
```

```
library(randomForest)
```

```
randomForest 4.7-1.2
Type rfNews() to see new features/changes/bug fixes.

Attaching package: 'randomForest'

The following object is masked from 'package:ggplot2':

    margin

The following object is masked from 'package:dplyr':

    combine
```

```
library(caret)
```

```
Loading required package: lattice

Attaching package: 'caret'

The following object is masked from 'package:purrr':

    lift
```

```
library(ggplot2)
library(report)
```

#load the dataset

```
gss_data<-read.csv("Random_Forest_Regression_Assignment.csv")
head(gss_data)
```

```
  House_ID  Location Bedrooms Bathrooms House_Size_m2 Plot_Size_m2
1    H0001  Roysambu        4         3         290.3        532.9
2    H0002 Kitengela        6         5         344.6        505.0
3    H0003     Ruaka        6         4         420.0        647.5
4    H0004     Runda        3         3         256.6        389.6
5    H0005  Kilimani        3         2         131.3        264.1
6    H0006  Kilimani        5         4         361.9        601.5
  House_Age_Years Distance_CBD_km Parking_Spaces Has_Garden Condition
1            16.0            11.9              2        Yes Excellent
2             1.3            10.0              3        Yes      Good
3             6.6            11.7              1        Yes      Fair
4             6.0             6.1              1        Yes      Good
5            11.0             8.6              0         No      Good
6             3.5             8.7              1        Yes      Good
  House_Price_Million_KES
1                   74.60
2                   62.30
3                   60.17
4                   76.31
5                   41.35
6                   80.16
```

#check structure

```
str(gss_data)
```

```
'data.frame':   800 obs. of  12 variables:
 $ House_ID              : chr  "H0001" "H0002" "H0003" "H0004" ...
 $ Location              : chr  "Roysambu" "Kitengela" "Ruaka" "Rund
a" ...
 $ Bedrooms              : int  4 6 6 3 3 5 4 3 5 2 ...
 $ Bathrooms             : int  3 5 4 3 2 4 4 2 4 2 ...
 $ House_Size_m2         : num  290 345 420 257 131 ...
 $ Plot_Size_m2          : num  533 505 648 390 264 ...
 $ House_Age_Years       : num  16 1.3 6.6 6 11 3.5 9.3 6.7 14.6 23.5
...
 $ Distance_CBD_km       : num  11.9 10 11.7 6.1 8.6 8.7 13.1 3.7 8.7
18.6 ...
 $ Parking_Spaces        : int  2 3 1 1 0 1 4 1 2 2 ...
 $ Has_Garden            : chr  "Yes" "Yes" "Yes" "Yes" ...
 $ Condition             : chr  "Excellent" "Good" "Fair" "Good" ...
 $ House_Price_Million_KES: num  74.6 62.3 60.2 76.3 41.4 ...
```

#check dimensions

```
dim(gss_data)
```

```
[1] 800  12
```

```
nrow(gss_data)
```

```
[1] 800
```

```
ncol(gss_data)
```

```
[1] 12
```

#check for missing values

```
colSums(is.na(gss_data))
```

```
               House_ID                Location                Bedroom
s
                      0                       0
0
              Bathrooms          House_Size_m2           Plot_Size_m
2
                      0                       0
0
       House_Age_Years       Distance_CBD_km         Parking_Space
s
                      0                       0
0
             Has_Garden               Condition House_Price_Million_KE
S
                      0                       0
0
```

check duplicate entries

```
sum(duplicated(gss_data))
```

```
[1] 0
```

#descriptive statistics

```
summary(gss_data)
```

```
      House_ID        Location      Bedrooms      Bathrooms     House_
Size_m2
 Length   :800   Length   :800   Min.   :2.00   Min.   :1.000   Min.
: 70.0
 N.unique :800   N.unique :  8   1st Qu.:3.00   1st Qu.:2.000   1st Q
u.:203.8
 N.blank  :  0   N.blank  :  0   Median :4.00   Median :3.000   Median
:263.2
 Min.nchar:  5   Min.nchar:  5   Mean   :3.88   Mean   :3.129   Mean
:265.1
 Max.nchar:  5   Max.nchar: 10   3rd Qu.:5.00   3rd Qu.:4.000   3rd Q
u.:325.8
                                 Max.   :6.00   Max.   :5.000   Max.
:420.0
  Plot_Size_m2   House_Age_Years  Distance_CBD_km  Parking_Spaces
 Min.   :180.0   Min.   : 1.000   Min.   : 1.000   Min.   :0.000
 1st Qu.:352.8   1st Qu.: 6.375   1st Qu.: 5.400   1st Qu.:1.000
 Median :456.4   Median :10.250   Median : 9.000   Median :1.000
 Mean   :464.3   Mean   :12.039   Mean   : 9.167   Mean   :1.405
 3rd Qu.:569.6   3rd Qu.:16.500   3rd Qu.:12.600   3rd Qu.:2.000
 Max.   :967.3   Max.   :35.000   Max.   :24.600   Max.   :4.000
     Has_Garden      Condition   House_Price_Million_KES
 Length   :800   Length   :800   Min.   : 25.00
 N.unique :  2   N.unique :  3   1st Qu.: 48.84
 N.blank  :  0   N.blank  :  0   Median : 62.47
 Min.nchar:  2   Min.nchar:  4   Mean   : 62.50
 Max.nchar:  3   Max.nchar:  9   3rd Qu.: 75.86
                                 Max.   :120.97
```

#explore the distribution of the target variables

```
summary(gss_data$House_Price_Million_KES)
```

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  25.00   48.84   62.47   62.50   75.86  120.97
```

#Creat a histogram

```
hist(gss_data$House_Price_Million_KES,
     main="Distribution of House Prices",
     xlab="House Price(KES)",
     ylab = "Frequency"
     )
```



#The distribution of house prices shows that most houses are concentrated around KES 40-80million,with a median price of approximately KES 60 million.The distribution is therefore fairly symmetric,although there are few high priced houses.

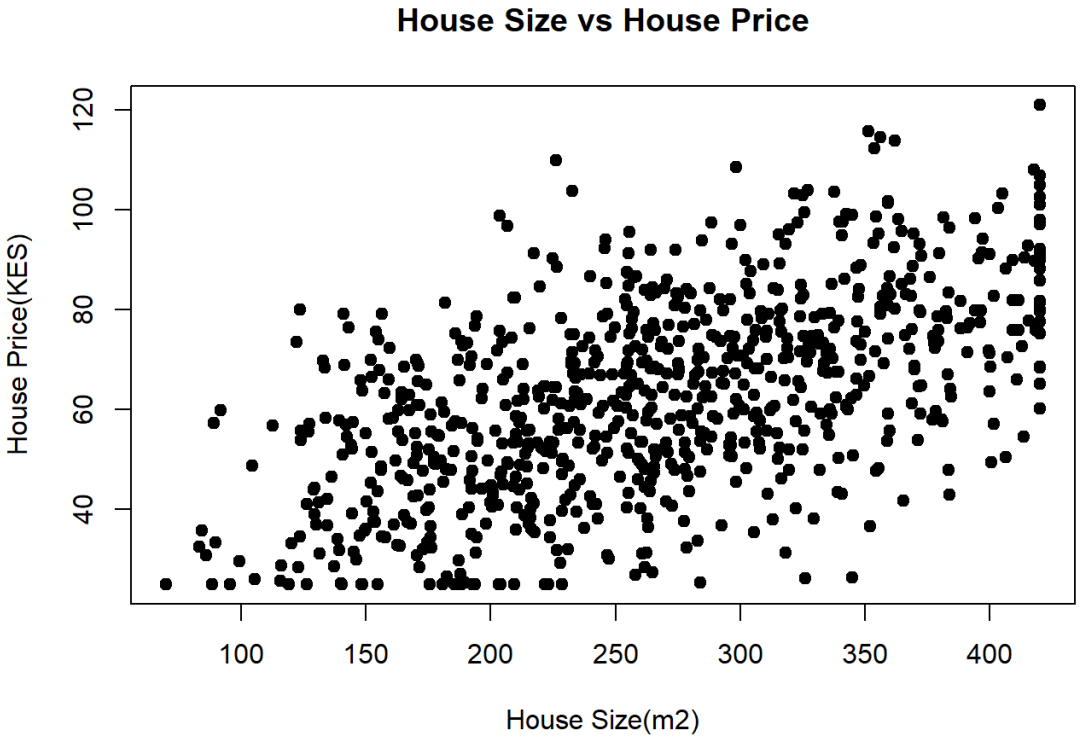
#Examine relationship between predictors and house price #numeric predictors # cORRELATION ANALYSIS

```
gss_selected<-gss_data|>
  select(House_Age_Years,House_Size_m2,Bedrooms,Plot_Size_m2,Parking_S
paces,Distance_CBD_km, House_Price_Million_KES)
cor(gss_selected)
```

```
                       House_Age_Years House_Size_m2     Bedrooms Plo
t_Size_m2
House_Age_Years              1.00000000   -0.01996723 -0.017094109
0.02436346
House_Size_m2               -0.01996723    1.00000000  0.843052683
0.52892818
Bedrooms                    -0.01709411    0.84305268  1.000000000
0.62834180
Plot_Size_m2                 0.02436346    0.52892818  0.628341799
1.00000000
Parking_Spaces               0.03507375    0.02195360  0.001507082
0.01855038
Distance_CBD_km              0.01883518   -0.04129293 -0.022604260  -
0.03502995
House_Price_Million_KES     -0.20320338    0.56176841  0.538630727
0.31392400
                       Parking_Spaces Distance_CBD_km House_Price_Mil
lion_KES
House_Age_Years            0.035073753      0.01883518              -
0.2032034
House_Size_m2              0.021953601     -0.04129293
0.5617684
Bedrooms                   0.001507082     -0.02260426
0.5386307
Plot_Size_m2               0.018550382     -0.03502995
0.3139240
Parking_Spaces             1.000000000      0.01760665
0.0799996
Distance_CBD_km            0.017606651      1.00000000              -
0.1913712
House_Price_Million_KES    0.079999600     -0.19137120
1.0000000
```

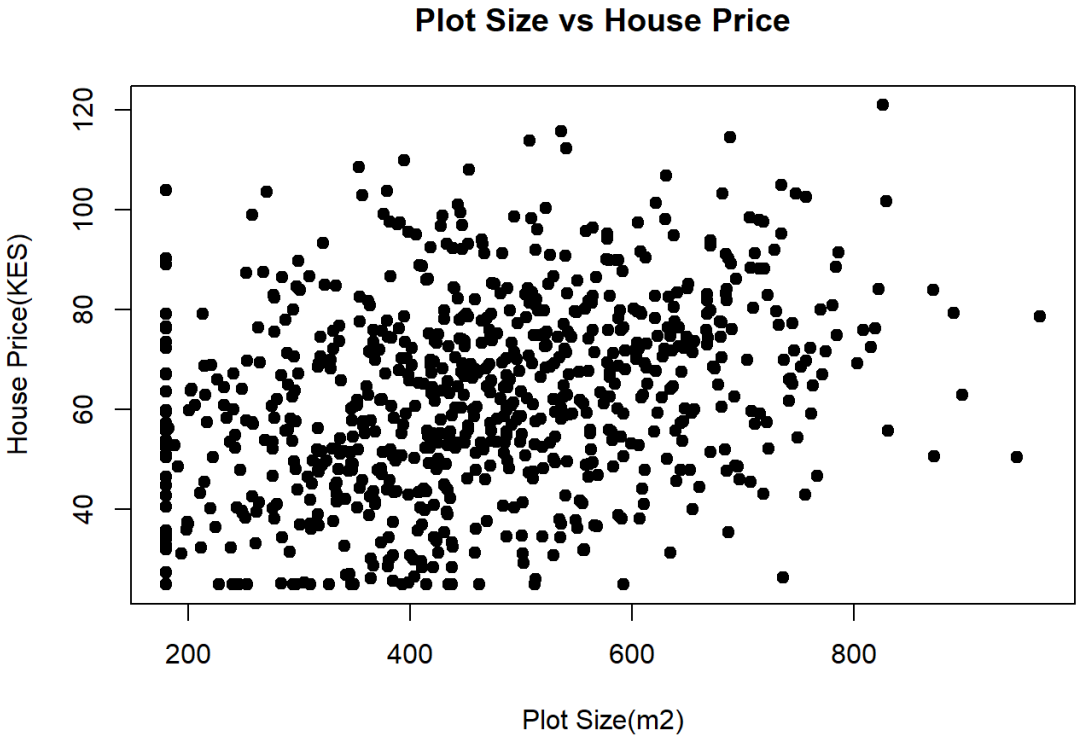
#House size vs house price

```
plot(gss_data$House_Size_m2,
     gss_data$House_Price_Million_KES,
     main="House Size vs House Price",
     xlab="House Size(m2)",
     ylab = "House Price(KES)",
     pch=19
     )
```



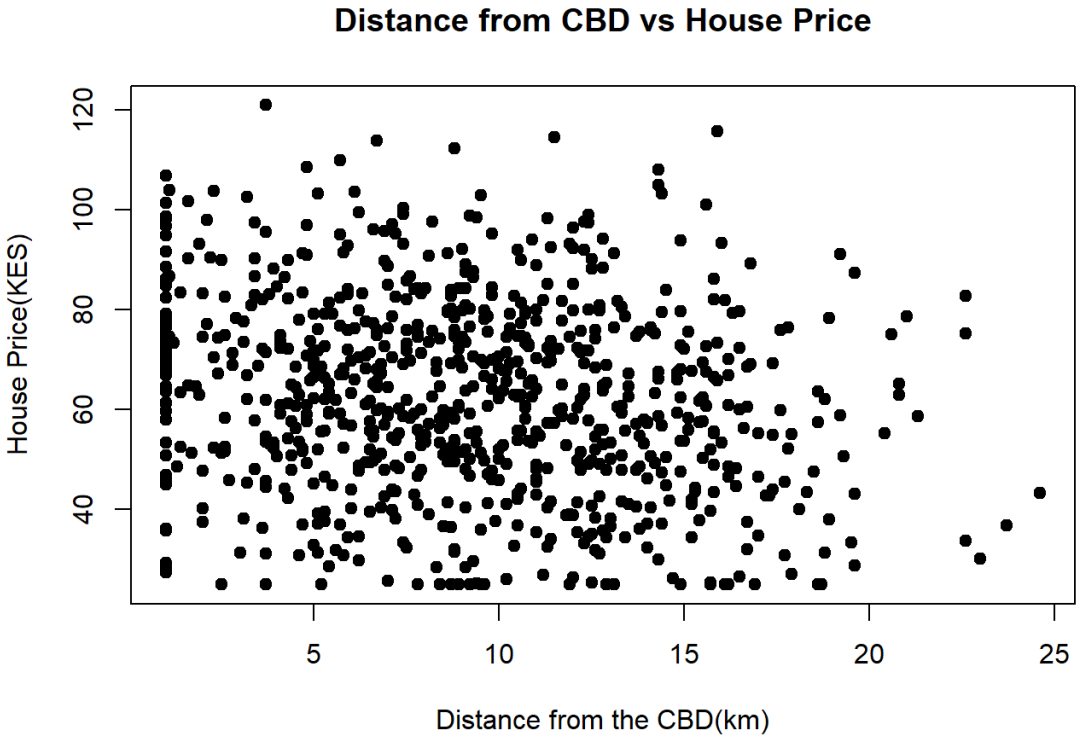
#Plot size vs house price

```
plot(gss_data$Plot_Size_m2,
     gss_data$House_Price_Million_KES,
     main="Plot Size vs House Price",
     xlab="Plot Size(m2)",
     ylab = "House Price(KES)",
     pch=19
     )
```



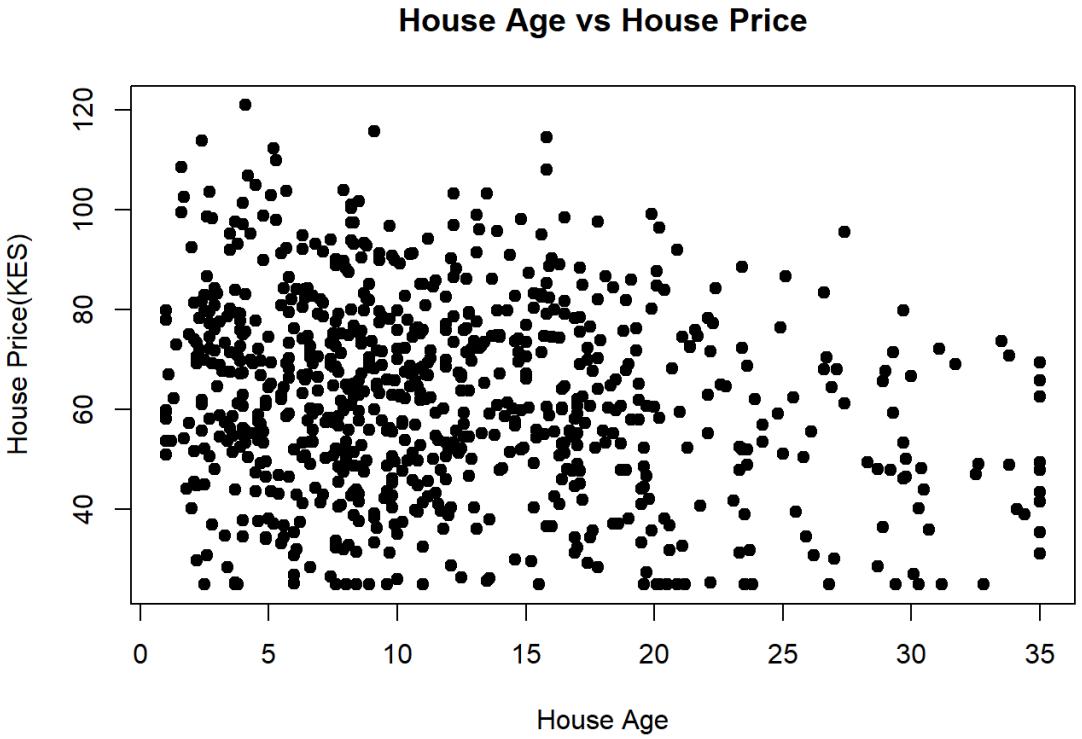
#Distance from CBD VS house price

```
plot(gss_data$Distance_CBD_km,
     gss_data$House_Price_Million_KES,
     main="Distance from CBD vs House Price",
     xlab="Distance from the CBD(km)",
     ylab = "House Price(KES)",
     pch=19
     )
```



#House age vs house price

```
plot(gss_data$House_Age_Years,
     gss_data$House_Price_Million_KES,
     main="House Age vs House Price",
    xlab="House Age",
     ylab = "House Price(KES)",
     pch=19
     )
```



#The scatter plots indicate that house size and plot size have a positive relationship with house prices,meaning that larger houses and plots generally tend to have higher prices.Distance from the CBD shows a weak negative relationship with house price,while house age also have a weak negative relationship.

#Convert categorical variables into numerical representation

```
gss_data$Location<-as.factor(
  gss_data$Location)
gss_data$Has_Garden<-as.factor(
  gss_data$Has_Garden)
gss_data$Condition<-as.factor(gss_data$Condition)
```

```
str(gss_data)
```

```
'data.frame':   800 obs. of  12 variables:
 $ House_ID              : chr  "H0001" "H0002" "H0003" "H0004" ...
 $ Location              : Factor w/ 8 levels "Kasarani","Kileleshw
a",..: 5 4 6 7 3 3 2 4 8 8 ...
 $ Bedrooms              : int  4 6 6 3 3 5 4 3 5 2 ...
 $ Bathrooms             : int  3 5 4 3 2 4 4 2 4 2 ...
 $ House_Size_m2         : num  290 345 420 257 131 ...
 $ Plot_Size_m2          : num  533 505 648 390 264 ...
 $ House_Age_Years       : num  16 1.3 6.6 6 11 3.5 9.3 6.7 14.6 23.5
...
 $ Distance_CBD_km       : num  11.9 10 11.7 6.1 8.6 8.7 13.1 3.7 8.7
18.6 ...
 $ Parking_Spaces        : int  2 3 1 1 0 1 4 1 2 2 ...
 $ Has_Garden            : Factor w/ 2 levels "No","Yes": 2 2 2 2 1 2
2 2 2 2 ...
 $ Condition             : Factor w/ 3 levels "Excellent","Fair",..:
1 3 2 3 3 3 1 3 1 2 ...
 $ House_Price_Million_KES: num  74.6 62.3 60.2 76.3 41.4 ...
```

#creat numerical representations

```
num_variables<-model.matrix(~Location+Has_Garden+Condition-1,
                            data=gss_data)
head(num_variables)
```

```
  LocationKasarani LocationKileleshwa LocationKilimani LocationKitenge
la
1                0                  0                0
0
2                0                  0                0
1
3                0                  0                0
0
4                0                  0                0
0
5                0                  0                1
0
6                0                  0                1
0
  LocationRoysambu LocationRuaka LocationRunda LocationSyokimau Has_Ga
rdenYes
1                1             0             0                0
1
2                0             0             0                0
1
3                0             1             0                0
1
4                0             0             1                0
1
5                0             0             0                0
0
6                0             0             0                0
1
  ConditionFair ConditionGood
1             0             0
2             0             1
3             1             0
4             0             1
5             0             1
6             0             1
```

#Divide the data into training and testing sets

```
set.seed(123)
sample_index<-sample(
  1:nrow(gss_data),
  size=0.8*nrow(gss_data)
)
train_data<-gss_data[sample_index,]
test_data<-gss_data[-sample_index,]
```

#check the dimention

```
dim(train_data)
```

```
[1] 640  12
```

```
dim(test_data)
```

```
[1] 160  12
```

#Build a random forest regression model

```
rf_model<-randomForest(
  House_Price_Million_KES~.,
  data=train_data,
  ntree=500,
  importance=TRUE
)
print(rf_model)
```

```

Call:
 randomForest(formula = House_Price_Million_KES ~ ., data = train_dat
a,      ntree = 500, importance = TRUE)
               Type of random forest: regression
                     Number of trees: 500
No. of variables tried at each split: 3

          Mean of squared residuals: 112.1107
                    % Var explained: 69.75
```

#Generate predictions for the test data

```
predicted_prices<-predict(rf_model,
                          newdata = test_data)
head(predicted_prices)
```

```
       1        3        9       17       22       25
66.24740 73.87851 72.64542 70.20676 59.22374 46.59951
```

```
head(test_data)
```

```
   House_ID Location Bedrooms Bathrooms House_Size_m2 Plot_Size_m2
1     H0001 Roysambu        4         3         290.3        532.9
3     H0003    Ruaka        6         4         420.0        647.5
9     H0009 Syokimau        5         4         368.4        580.1
17    H0017 Roysambu        5         4         327.4        814.9
22    H0022 Kilimani        3         1         191.7        356.9
25    H0025 Kasarani        3         3         174.5        437.6
   House_Age_Years Distance_CBD_km Parking_Spaces Has_Garden Condition
1             16.0            11.9              2        Yes Excellent
3              6.6            11.7              1        Yes      Fair
9             14.6             8.7              2        Yes Excellent
17            21.4             1.0              0        Yes      Good
22            10.6             2.7              1         No      Good
25             9.1            19.5              1         No Excellent
   House_Price_Million_KES
1                    74.60
3                    60.17
9                    82.70
17                   72.44
22                   45.89
25                   33.32
```

#Evaluate the model

```
rmse<-RMSE(predicted_prices,test_data$House_Price_Million_KES)
mae<-MAE(predicted_prices,test_data$House_Price_Million_KES)
r2<-R2(predicted_prices,test_data$House_Price_Million_KES)
cat("RMSE:",rmse,"\n")
```

```
RMSE: 10.78676
```

```
cat("MAE:",mae,"\n")
```

```
MAE: 8.839857
```

```
cat("R-squared:",r2,"\n")
```

```
R-squared: 0.7880161
```

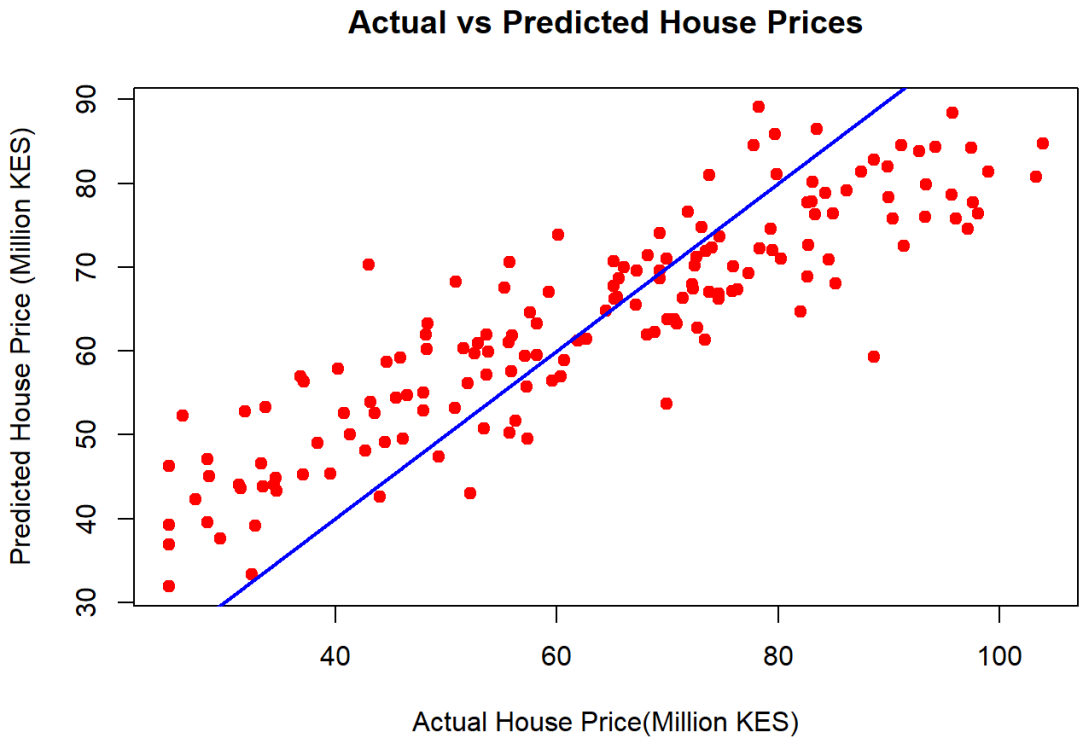
#compare actual and predicted house prices

```
comparison<-data.frame(
  Actual_price=test_data$House_Price_Million_KES,
  predicted_price=predicted_prices
)
head(comparison,20)
```

```
   Actual_price predicted_price
1         74.60        66.24740
3         60.17        73.87851
9         82.70        72.64542
17        72.44        70.20676
22        45.89        59.22374
25        33.32        46.59951
27        59.32        67.02213
28        25.00        31.89296
32        68.24        71.44826
42        40.83        52.56872
43        65.43        66.47309
44        34.71        43.35661
58        91.10        84.53542
60        34.52        44.03420
62        26.27        52.26753
63        45.49        54.41653
70        57.40        49.51494
77        37.22        56.35599
82        53.42        50.77609
86        52.62        59.68303
```

#plot actual vs predicted values

```
plot(
  test_data$House_Price_Million_KES,
  predicted_prices,
  main="Actual vs Predicted House Prices",
  xlab="Actual House Price(Million KES)",
  ylab="Predicted House Price (Million KES)",
  pch=19,
  col="red"
)
abline(0,1,col="blue",lwd=2)
```

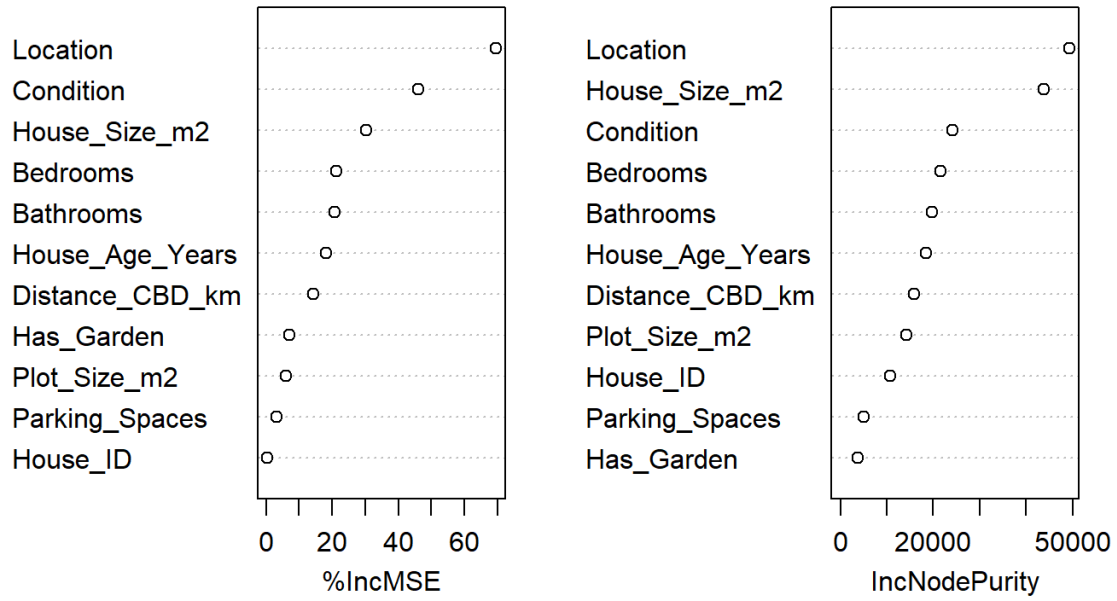


#Examine variable importance

```
importance(rf_model)
```

```
                  %IncMSE IncNodePurity
House_ID         0.2484834     10687.371
Location        69.6648491     49417.093
Bedrooms        21.3684051     21505.726
Bathrooms       20.6216358     19833.932
House_Size_m2   30.2669056     43891.478
Plot_Size_m2     5.9538736     14226.023
House_Age_Years 18.2646941     18416.227
Distance_CBD_km 14.2442858     15843.540
Parking_Spaces   3.1859348      5074.209
Has_Garden       7.0572012      3782.330
Condition       46.0272115     24105.517
```

```
varImpPlot(rf_model,
           main="Variable importance in Random Forest Model")
```




```
importance_table<-data.frame(
  variable=rownames(importance(rf_model)),
  importance(rf_model)
)
importance_table
```

```
                      variable   X.IncMSE IncNodePurity
House_ID               House_ID  0.2484834     10687.371
Location               Location 69.6648491     49417.093
Bedrooms               Bedrooms 21.3684051     21505.726
Bathrooms             Bathrooms 20.6216358     19833.932
House_Size_m2     House_Size_m2 30.2669056     43891.478
Plot_Size_m2       Plot_Size_m2  5.9538736     14226.023
House_Age_Years House_Age_Years 18.2646941     18416.227
Distance_CBD_km Distance_CBD_km 14.2442858     15843.540
Parking_Spaces   Parking_Spaces  3.1859348      5074.209
Has_Garden           Has_Garden  7.0572012      3782.330
Condition             Condition 46.0272115     24105.517
```

#identify the most important predictors

```
importance_table
```

```
                      variable   X.IncMSE IncNodePurity
House_ID               House_ID  0.2484834     10687.371
Location               Location 69.6648491     49417.093
Bedrooms               Bedrooms 21.3684051     21505.726
Bathrooms             Bathrooms 20.6216358     19833.932
House_Size_m2     House_Size_m2 30.2669056     43891.478
Plot_Size_m2       Plot_Size_m2  5.9538736     14226.023
House_Age_Years House_Age_Years 18.2646941     18416.227
Distance_CBD_km Distance_CBD_km 14.2442858     15843.540
Parking_Spaces   Parking_Spaces  3.1859348      5074.209
Has_Garden           Has_Garden  7.0572012      3782.330
Condition             Condition 46.0272115     24105.517
```

```
head(importance_table,5)
```

```
                   variable   X.IncMSE IncNodePurity
House_ID           House_ID  0.2484834      10687.37
Location           Location 69.6648491      49417.09
Bedrooms           Bedrooms 21.3684051      21505.73
Bathrooms         Bathrooms 20.6216358      19833.93
House_Size_m2 House_Size_m2 30.2669056      43891.48
```